

DATABASE SPECIFICATION - AIMS PROJECT

Table: Product

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		productID	Integer	Yes	ID, auto increment
2			title	VARCHAR(255)	Yes	Product title
3			category	VARCHAR(45)	Yes	Media category
4			description	TEXT	No	General description
5			barcode	VARCHAR(50)	No	Barcode
6			length	Double	No	Length dimension
7			width	Double	No	Width dimension
8			height	Double	No	Height dimension
9			weight	Double	No	Weight
10			originalValue	Double	Yes	Original price (excl. 10% VAT)
11			currentPrice	Double	Yes	Current selling price (excl. 10% VAT)
12			quantityInStock	Integer	Yes	Quantity in stock
13			status	VARCHAR(20)	No	ACTIVE, DEACTIVATED
14			imageUrl	VARCHAR(255)	No	Image URL path

Table: PrintableProduct

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x	x	productID	Integer	Yes	ID, ref to Product
2			type	VARCHAR(45)	Yes	BOOK or NEWSPAPER
3			publisher	VARCHAR(255)	No	Publisher name
4			publishDate	DATETIME	No	Date of publication
5			language	VARCHAR(45)	No	Language

Table: Book

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x	x	productID	Integer	Yes	ID, ref to PrintableProduct
2			coverType	VARCHAR(45)	No	Cover type
3			numberOfPage	Integer	No	Number of pages
4			bookCategory	VARCHAR(100)	No	Book category

Table: Author

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		authorID	Integer	Yes	ID, auto increment
2			name	VARCHAR(255)	No	Author name

Table: Book_Author (N-N)

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x	x	productID	Integer	Yes	Ref to Book.productID
2	x	x	authorID	Integer	Yes	Ref to Author.authorID

Table: Newspaper

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x	x	productID	Integer	Yes	ID, ref to PrintableProduct
2			editorInChief	VARCHAR(255)	No	Editor in chief
3			issueNumber	VARCHAR(50)	No	Issue number
4			publicationFrequency	VARCHAR(50)	No	Publication frequency
5			issn	VARCHAR(50)	No	ISSN code
6			sections	TEXT	No	Politics, Business, etc.

Table: DiscProduct

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x	x	productID	Integer	Yes	ID, ref to Product
2			type	VARCHAR(45)	Yes	CD or DVD
3			genre	VARCHAR(100)	No	Genre
4			language	VARCHAR(45)	No	Language
5			releasedDate	DATETIME	No	Release date

Table: CD

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x	x	productID	Integer	Yes	ID, ref to DiscProduct
2			artists	VARCHAR(255)	No	Artists
3			recordLabel	VARCHAR(255)	No	Record label
4			musicType	VARCHAR(45)	No	Music type

Table: Track

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		trackID	Integer	Yes	ID, auto increment
2			title	VARCHAR(255)	No	Track title
3			length	Integer	No	Duration in seconds

Table: CD_Track (N-N)

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x	x	productID	Integer	Yes	Ref to CD.productID
2	x	x	trackID	Integer	Yes	Ref to Track.trackID

Table: DVD

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x	x	productID	Integer	Yes	ID, ref to DiscProduct
2			discType	VARCHAR(45)	No	Disc type
3			director	VARCHAR(255)	No	Director
4			runtime	Integer	No	Runtime
5			studio	VARCHAR(255)	No	Studio
6			subtitles	VARCHAR(255)	No	Subtitles

Table: Role

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		roleID	Integer	Yes	ID, auto increment
2			name	VARCHAR(50)	No	ADMIN, PRODUCT_MANAGER

Table: User

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		userID	Integer	Yes	ID, auto increment
2			name	VARCHAR(255)	No	User name
3			email	VARCHAR(255)	No	Email (Unique)
4			password	VARCHAR(255)	No	Password
5			status	VARCHAR(20)	No	ACTIVE, BLOCKED

Table: User_Role (N-N)

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x	x	userID	Integer	Yes	Ref to User.userID
2	x	x	roleID	Integer	Yes	Ref to Role.roleID

Table: ActionLogHistory

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		ActionLogHistoryID	Integer	Yes	ID, auto increment
2		x	performedBy	Integer	No	Ref to User.userID
3			type	VARCHAR(50)	No	Action type
4			description	TEXT	No	Action description
5			createdAt	DATETIME	No	Creation timestamp

Table: ProductHistoryLog

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		ProductHistoryLogID	Integer	Yes	ID, auto increment
2		x	productID	Integer	No	Ref to Product.productID
3		x	performedBy	Integer	No	Ref to User.userID
4			updatedBy	VARCHAR(255)	No	Updater name
5			action	VARCHAR(50)	No	ADD, EDIT, DELETE
6			reason	TEXT	No	Required for manual stock change
7			timestamp	DATETIME	No	Timestamp
8			description	TEXT	No	Description

Table: Order

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		orderID	Integer	Yes	ID, auto increment
2			shippingFee	Double	No	Shipping fee
3			status	VARCHAR(50)	No	Order status
4			createdAt	DATETIME	No	Creation timestamp

Table: OrderItem

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		orderItemID	Integer	Yes	ID, auto increment
2		x	orderID	Integer	No	Ref to Order.orderID
3		x	productID	Integer	No	Ref to Product.productID
4			quantity	Integer	No	Quantity
5			unitPrice	Double	No	Unit price

Table: DeliveryInfo

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		deliveryID	Integer	Yes	ID, auto increment
2		x	orderID	Integer	Yes	Ref to Order.orderID (Unique)
3			receiverName	VARCHAR(255)	No	Receiver name
4			email	VARCHAR(255)	No	Email
5			phoneNumber	VARCHAR(20)	No	Phone number
6			address	VARCHAR(255)	No	Address
7			province	VARCHAR(100)	No	Province
8			delivery_notes	TEXT	No	Delivery notes

Table: Invoice

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		invoiceID	Integer	Yes	ID, auto increment
2		x	orderID	Integer	Yes	Ref to Order.orderID (Unique)
3			totalExcludeVAT	Double	No	Total excl. VAT
4			totalIncludeVAT	Double	No	Total incl. 10% VAT
5			shippingFee	Double	No	Shipping fee
6			totalPayment	Double	No	VAT Price + Shipping Fee
7			createdAt	DATETIME	No	Creation timestamp

Table: PaymentTransaction

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		transactionID	Integer	Yes	ID, auto increment
2		x	invoiceID	Integer	No	Ref to Invoice.invoiceID
3			method	VARCHAR(45)	No	VIETQR, PAYPAL
4			amount	Double	No	Amount
5			transactionContent	TEXT	No	Transaction content
6			createdAt	DATETIME	No	Creation timestamp

Table: PaypalTransaction

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		paypalTransactionID	Integer	Yes	ID, auto increment
2		x	transactionID	Integer	Yes	Ref to PaymentTransaction.transactionID (Unique)
3			paypalOrderID	VARCHAR(100)	No	PayPal Order ID
4			paypalCaptureID	VARCHAR(100)	No	PayPal Capture ID
5			payerID	VARCHAR(100)	No	Payer ID
6			amount	Double	No	Amount
7			status	VARCHAR(50)	No	Status